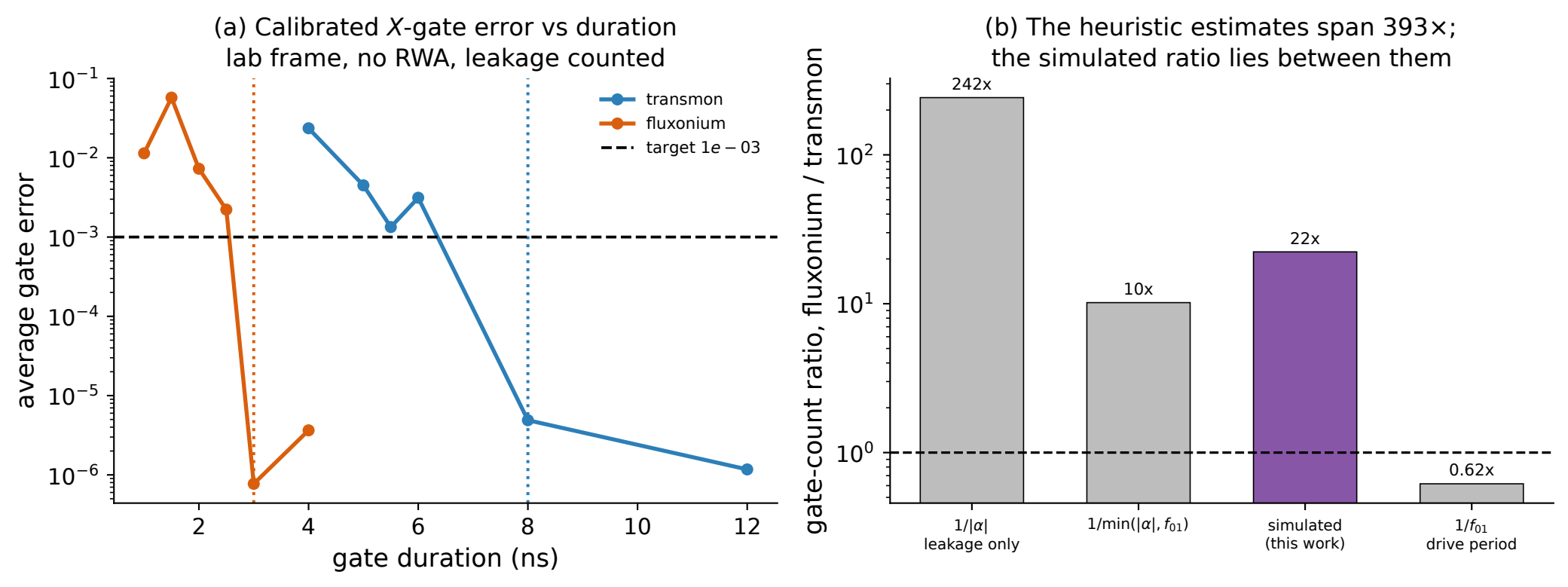

(a) Calibrated X-gate error vs duration
lab frame, no RWA, leakage counted
average gate error
gate duration (ns)
transmon
fluxonium
target 1e − 03
(b) The heuristic estimates span 393×;
the simulated ratio lies between them
gate-count ratio, fluxonium / transmon
242x
10x
22x
0.62x
1/|α|
leakage only
1/min(|α|, f01)
simulated
(this work)
1/f01
drive period